

Chapter 20: Electric Current, Resistance, & Ohm's Law

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PHY201
Lecture 20



Part I

Electric Current

Outline of Part I: Electric Current

Current

Outline of Part I: Electric Current

Current

Electricity and Fluid Dynamics

Fluid Dynamics

► Current: **flow of mass**

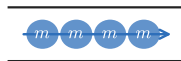
► $\dot{m} = \frac{\Delta m}{\Delta t}$

Electricity

► Current: **flow of charge**

► $I = \frac{\Delta Q}{\Delta t}$

Fluid



$$\dot{m} = \frac{\Delta m}{\Delta t}$$

mass flow
[kg/s]

Electric



$$I = \frac{\Delta Q}{\Delta t}$$

charge flow
[C/s = A]

Electric Current

Electric Current — SI Unit: ampere (A = C/s)

Electric current I is the net rate at which charge flows past a cross-section:

$$I = \frac{\Delta Q}{\Delta t}$$

- ▶ ΔQ = charge that passes a cross-section [C]
- ▶ Δt = elapsed time [s]
- ▶ $[I] = \text{ampere (A)} = 1 \frac{\text{C}}{\text{s}}$

Named after André-Marie Ampère (1775–1836), French physicist and mathematician.

Conventional Current Direction

Convention

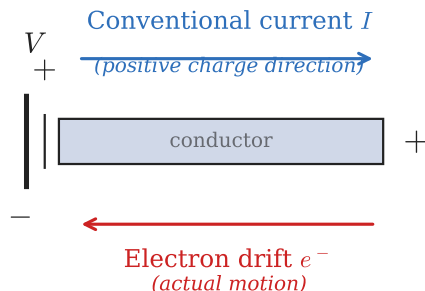
Conventional current flows from the **positive** to the **negative** terminal — the direction a *positive* charge would move.

Reality in metals

Electrons (negative charges) drift in the *opposite* direction. We still draw current arrows as if positive charges flow.

Historical note

The convention was established before electrons were discovered.



Drift Velocity vs. Signal Speed

Drift velocity

Average speed of electrons through a metal:

$$v_d \sim 10^{-4} \frac{\text{m}}{\text{s}}$$

Very slow — electrons zigzag constantly due to collisions.

Signal speed

Speed at which the *electrical influence* propagates:

$$v_{\text{signal}} \sim 10^8 \frac{\text{m}}{\text{s}}$$

Nearly the speed of light.

Garden-hose analogy

When you turn on a tap, water flows from the far end *immediately* — even though the same water molecules travel slowly. The influence propagates far faster than the drift speed of molecules.

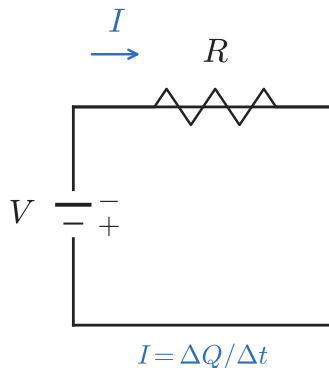
Example: Electric Current

Find I (cp2e Ex. 20.1)

A car battery delivers $\Delta Q = 720 \text{ C}$ in $\Delta t = 4.00 \text{ s}$. Find the current.

$$I = \frac{\Delta Q}{\Delta t} = \frac{720 \text{ C}}{4.00 \text{ s}}$$

$$I = 180 \text{ A}$$



Check Your Understanding

Can a wire carry current even though it is electrically neutral overall?

Come to lecture to see the answer.

Check Your Understanding

Why do you need *two* conducting paths to operate an electrical device (e.g., two wires to a lamp), rather than just one?

Come to lecture to see the answer.

Part II

Ohm's Law & Resistance

Outline of Part II: Ohm's Law & Resistance

Ohm's Law, Resistance, & Simple Circuits

Resistance & Resistivity

Outline of Part II: Ohm's Law & Resistance

Ohm's Law, Resistance, & Simple Circuits

Resistance & Resistivity

Ohm's Law

Two proportionalities

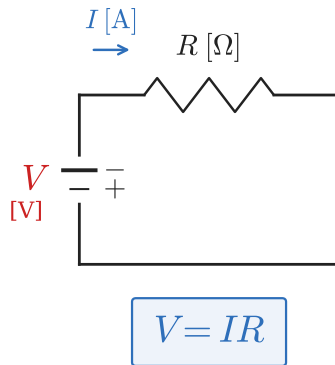
- ▶ $I \propto V$ (more voltage $\rightarrow$ more current)
- ▶ $I \propto \frac{1}{R}$ (more resistance $\rightarrow$ less current)

Combined — **Ohm's Law** — SI Unit: ohm
($\Omega = \frac{V}{A}$):

$$V = IR$$

Named after

Georg Simon Ohm (1789–1854), German physicist.



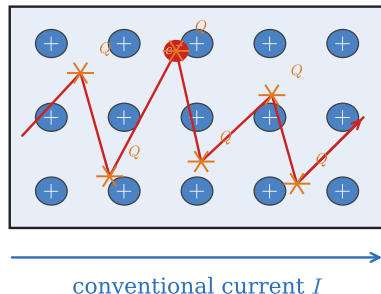
What is Resistance?

Physical origin

- ▶ Electrons collide with atoms as they drift through a conductor.
- ▶ Collisions convert kinetic energy to **thermal energy** (heat).
- ▶ More collisions $\rightarrow$ higher resistance.

Ohmic vs. non-ohmic

- ▶ **Ohmic:** R constant (copper, aluminum)
- ▶ **Non-ohmic:** R varies (diodes, transistors, light bulbs)



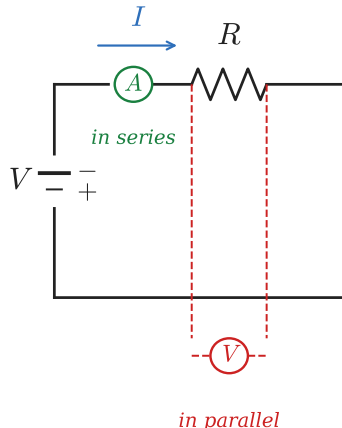
Measuring Circuits: Voltmeter & Ammeter

Voltmeter

- ▶ Measures potential difference ΔV
- ▶ Connected **in parallel** with the element
- ▶ Very *high* resistance (draws ≈ 0 current)

Ammeter

- ▶ Measures current I
- ▶ Connected **in series** with the element
- ▶ Very *low* resistance (minimal voltage drop)



Memory aid

Check Your Understanding

Does the current change in magnitude as it passes through a resistor?

Come to lecture to see the answer.

Check Your Understanding

If you double the voltage across a resistor while keeping R constant, what happens to the current? What happens to the power?

Come to lecture to see the answer.

Outline of Part II: Ohm's Law & Resistance

Ohm's Law, Resistance, & Simple Circuits

Resistance & Resistivity

Resistivity ρ vs. Resistance R

Resistivity ρ

- ▶ *Intrinsic* material property
- ▶ Does *not* depend on shape or size
- ▶ Units: $\Omega \cdot \text{m}$
- ▶ Determines how strongly a material opposes current

Resistance R

- ▶ Depends on *both* material *and* geometry
- ▶ Units: Ω
- ▶ Changes with length and cross-sectional area

Analogy

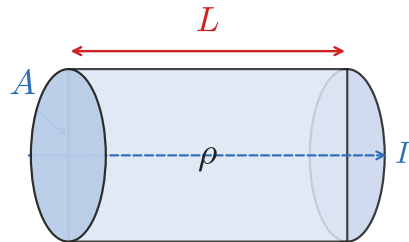
Resistivity $\sim$ a material's *density*. Resistance $\sim$ the *weight* of a specific object (depends on density AND size).

Resistance & Geometry

Resistance Formula — SI Unit: ohm (Ω)

$$R = \frac{\rho L}{A}$$

- ▶ ρ = resistivity [$\Omega \cdot \text{m}$]
- ▶ L = length [m]
- ▶ A = cross-sectional area [m^2]



$$R = \frac{\rho L}{A}$$

Physical intuition

- ▶ Longer L : more collisions $\rightarrow$ larger R
- ▶ Larger A : more parallel paths $\rightarrow$ smaller

Resistivity of Common Materials

Conductors (small ρ)

- ▶ Silver: $\rho = 1.59 \times 10^{-8} \Omega \cdot \text{m}$ (best conductor)
- ▶ Copper: $\rho = 1.72 \times 10^{-8} \Omega \cdot \text{m}$ (most common wire)
- ▶ Aluminum: $\rho = 2.65 \times 10^{-8} \Omega \cdot \text{m}$

Semiconductors (intermediate ρ)

- ▶ Silicon: $\rho \approx 2300 \Omega \cdot \text{m}$ (depends heavily on impurities/doping)

Insulators (large ρ)

- ▶ Rubber: $\rho \sim 10^{13} \Omega \cdot \text{m}$
- ▶ Quartz: $\rho \sim 7.5 \times 10^{17} \Omega \cdot \text{m}$

Temperature Dependence of Resistivity

Temperature coefficient of resistivity

$$\rho(T) = \rho_0(1 + \alpha \Delta T)$$

- ▶ ρ_0 = resistivity at reference temperature (usually 20 °C)
- ▶ α = temperature coefficient of resistivity [$\frac{1}{^\circ\text{C}}$]
- ▶ $\Delta T = T - T_0$

Metals ($\alpha > 0$)

Resistance *increases* with temperature.
More vibration → more collisions.

Semiconductors ($\alpha < 0$)

Resistance *decreases* with temperature.
Heat frees more charge carriers.

Check Your Understanding

Two wires have the same length and the same resistance. One is made of aluminum ($\rho_{\text{Al}} = 2.65 \times 10^{-8} \Omega \cdot \text{m}$) and the other of copper ($\rho_{\text{Cu}} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$). Which wire has a larger diameter?

Come to lecture to see the answer.

Check Your Understanding

A light bulb's resistance is much higher when it is hot (glowing) than when it is cold (just switched on). Is the bulb's filament a metal or a semiconductor? How do you know?

Come to lecture to see the answer.

Part III

Power, Energy, & AC/DC

Outline of Part III: Power, Energy, & AC/DC

Electric Power & Energy

Alternating Current vs. Direct Current

Outline of Part III: Power, Energy, & AC/DC

Electric Power & Energy

Alternating Current vs. Direct Current

Electric Power

Electric Power — SI Unit: watt ($W = J/s$)

Substituting $V = IR$ into $P = E/t$ gives three equivalent forms:

$$P = IV = \frac{V^2}{R} = I^2 R = \frac{E}{t}$$

Which form to use?

- ▶ Know I and V ? Use $P = IV$.
- ▶ Know V and R ? Use $P = V^2/R$.
- ▶ Know I and R ? Use $P = I^2 R$.

Key insight

$P = IV$ always holds. The other two forms follow from substituting $V = IR$.

Energy & Electric Bills

Energy consumed

From $P = E/t$:

$$E = P t$$

- ▶ SI unit: joule (J)
- ▶ Utility billing unit: **kilowatt-hour** (kWh)
- ▶ $1 \text{ kW h} = 3.6 \times 10^6 \text{ J}$

Cost formula

$$\text{Cost} = P \cdot t \cdot \text{rate} \quad \left(\text{rate in } \frac{\$}{\text{kWh}} \right)$$

Example: Electric Power & Cost

Whiteboard Example (cp2e Ex. 20.7)

A 1500 W space heater runs for 8 hours. If the electricity rate is \$0.12/kWh, find the cost.

Example: Electric Power & Cost

Whiteboard Example (cp2e Ex. 20.7)

A 1500 W space heater runs for 8 hours. If the electricity rate is \$0.12/kWh, find the cost.

$$\text{Cost} = P \cdot t \cdot \text{rate}$$

$$= 1500 \text{ W} \cdot 8 \text{ h} \cdot \frac{\$0.12}{\text{kW h}} \cdot \frac{1 \text{ kW}}{1000 \text{ W}}$$

$$\boxed{\text{Cost} = \$1.44}$$

25-W vs. 60-W Light Bulb

Both rated for 120 V circuits

- ▶ Power rating = rate of energy consumption.
- ▶ The 60-W bulb needs *more* current than the 25-W bulb.

Which has smaller resistance?

Using $P = V^2/R$:

$$R_{25} = \frac{(120 \text{ V})^2}{25 \text{ W}} = 576 \Omega \quad R_{60} = \frac{(120 \text{ V})^2}{60 \text{ W}} = 240 \Omega$$

The 60-W bulb has **smaller** resistance.

Key insight

For a fixed voltage, *smaller* resistance $\rightarrow$ *more* current $\rightarrow$ *more* power dissipated.

Check Your Understanding

Two resistors are connected to the same voltage source. Resistor A has twice the resistance of resistor B. Which dissipates more power?

Come to lecture to see the answer.

Check Your Understanding

Incandescent light bulbs often burn out at the moment they are switched on rather than after extended use. Using what you know about temperature and resistance, explain why.

Come to lecture to see the answer.

Outline of Part III: Power, Energy, & AC/DC

Electric Power & Energy

Alternating Current vs. Direct Current

DC vs. AC

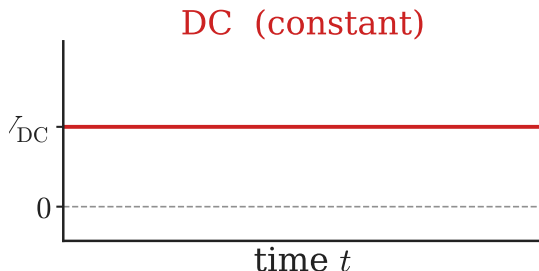
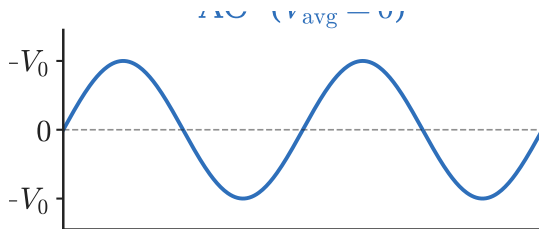
Direct Current (DC)

- ▶ Current in one direction; constant V
- ▶ Sources: batteries, solar cells

Alternating Current (AC)

- ▶ Current reverses periodically; V oscillates
- ▶ Source: wall outlets (60 Hz in the US)

Why AC for power transmission?



AC Voltage and Current

Time-varying voltage and current

$$V(t) = V_0 \sin(2\pi ft)$$

Current in a resistive load: $I(t) = I_0 \sin(2\pi ft)$, where $I_0 = V_0/R$.

Average values over a full cycle

$V_{\text{avg}} = 0$ and $I_{\text{avg}} = 0$ because the positive and negative halves cancel exactly.

Implication

We *cannot* use average values to compute power. We need **RMS** values.

RMS Values

Root Mean Square (RMS)

$$I_{\text{RMS}} = \frac{I_0}{\sqrt{2}}$$

$$V_{\text{RMS}} = \frac{V_0}{\sqrt{2}}$$

The RMS value is the equivalent DC value that produces the *same average power*.

Average power delivered to a resistor

$$P_{\text{ave}} = \frac{1}{2} I_0 V_0 = I_{\text{RMS}} V_{\text{RMS}}$$

US household outlets

$V_{\text{RMS}} = 120 \text{ V}$, so $V_0 = 120\sqrt{2} \approx 170 \text{ V}$ (peak)

Check Your Understanding

Why is the average of an AC voltage zero, yet the average power delivered to a resistor is *not* zero?

Come to lecture to see the answer.

Check Your Understanding

A household outlet is rated at $V_{\text{RMS}} = 120 \text{ V}$. What is the peak voltage V_0 ?

Come to lecture to see the answer.

Part IV

Electric Hazards & Bioelectricity

Outline of Part IV: Electric Hazards & Bioelectricity

Electric Hazards & the Human Body

Nerve Conduction & Electrocardiograms

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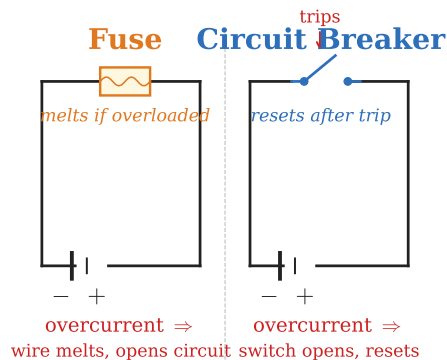
Two Types of Electrical Hazards

Thermal hazards

- ▶ $P = I^2 R$ heating $\rightarrow$ fires, melted insulation
- ▶ Cause: circuit overloads, short circuits
- ▶ Protection: **fuses** & **circuit breakers**

Shock hazards

- ▶ Dangerous current through the body
- ▶ Risk depends on current, path through body, duration
- ▶ Can cause ventricular fibrillation



Current Effects on the Human Body

Physiological effects

Current	Effect
1 mA	Threshold of sensation
5 mA	Maximum harmless current
10–20 mA	Muscle contraction; “cannot let go”; breathing may stop
50 mA	Onset of pain
100–300 mA	Ventricular fibrillation — often fatal
6 A	Sustained contraction; may defibrillate

Key fact

It is **current** that kills, not voltage — but voltage drives the current through the body's resistance.

Skin Resistance & Shock Danger

Dry skin vs. wet skin

- ▶ Dry skin: $R \approx 200\,000\ \Omega$ (protective)
- ▶ Wet skin: $R \approx 1000\ \Omega$ ($200\times$ less protection!)

Example: touching a 120 V outlet

- ▶ Dry: $I = 120\text{ V}/200\,000\ \Omega = 0.6\text{ mA}$ (usually harmless)
- ▶ Wet: $I = 120\text{ V}/1000\ \Omega = 120\text{ mA}$ (potentially fatal!)

Safety rules

Never use electrical appliances near water. Never handle live wires with wet hands.

Microshock Hazard

When skin resistance is bypassed

IV lines, catheters, or surgical probes deliver current *directly* to internal tissue.

The body's internal resistance is only $\sim 500\ \Omega$ — much lower than skin.

Only $\sim 10\ \mu\text{A}$ of current *directly to the heart* can cause ventricular fibrillation.

Clinical implication

Hospital operating rooms and ICUs require **isolated electrical systems** and **ground fault interrupters** (GFI/GFCI) to prevent microshock to patients.

Check Your Understanding

A person is in a bathtub when a hair dryer falls into the water. Why is this so dangerous, even if the person is not touching the dryer directly?

Come to lecture to see the answer.

Check Your Understanding

Electricians often work with one hand behind their back or in their pocket. Why?

Come to lecture to see the answer.

Check Your Understanding

A short circuit occurs when a low-resistance path accidentally connects two points at different potentials. Why is this so dangerous from a thermal hazard standpoint?

Come to lecture to see the answer.

Outline of Part IV: Electric Hazards & Bioelectricity

Electric Hazards & the Human Body

Nerve Conduction & Electrocardiograms

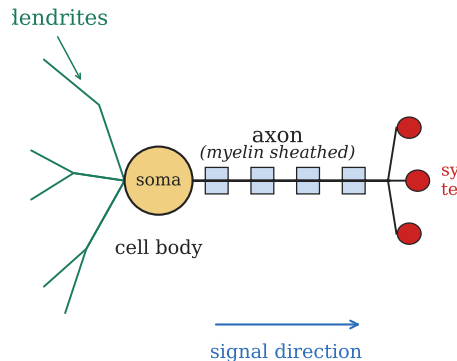
Bioelectricity: Electricity in the Body

Electrical signals everywhere

- ▶ Nerves carry electrical impulses to sense, move, and think.
- ▶ Heart muscle is driven and coordinated electrically.

Neuron structure

- ▶ **Dendrites:** receive signals from other neurons
- ▶ **Axon:** transmit signal to the next cell
- ▶ **Synapse:** junction; neurotransmitters cross the gap



Resting Membrane Potential

Resting potential $\approx -90 \text{ mV}$

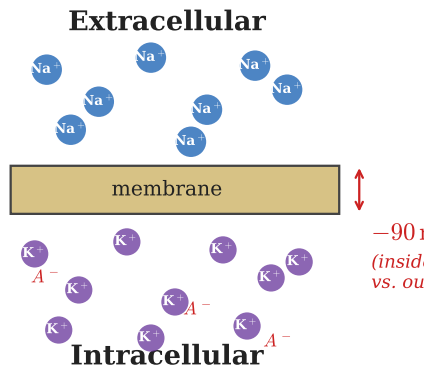
Inside of the cell is negative relative to outside at rest.

- ▶ Na^+ concentration higher *outside*
- ▶ K^+ concentration higher *inside*
- ▶ Maintained by ion pumps (using ATP)

Action potential

When stimulated above threshold:

- ▶ Na^+ rushes in $\rightarrow$ **depolarization** (voltage spikes positive)
- ▶ K^+ flows out $\rightarrow$ **repolarization** (returns to -90 mV)



Nerve Impulse Propagation

How signals travel along a nerve

Depolarization of one membrane segment triggers adjacent segments → a wave of depolarization travels at $\sim 1 \frac{\text{m}}{\text{s}}$ in unmyelinated nerves.

Myelin sheaths (saltatory conduction)

Insulating myelin around axons forces the signal to *jump* between exposed gaps (nodes of Ranvier):

- ▶ Speed increases to $\sim 100 \frac{\text{m}}{\text{s}}$
- ▶ Prevents cross-talk between adjacent nerves

Multiple sclerosis (MS)

Myelin sheaths are damaged, slowing or blocking nerve signals and causing loss of muscle control.

Electrocardiogram (ECG / EKG)

What an ECG records

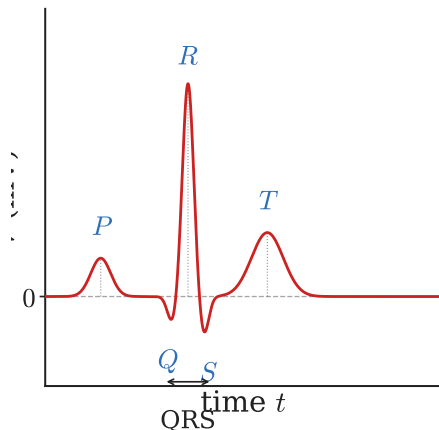
Skin electrodes detect the voltages produced as heart muscle cells depolarize and repolarize.

Key features of a normal ECG

- ▶ **P wave:** atria depolarize and contract
- ▶ **QRS complex:** ventricles depolarize (main pumping beat)
- ▶ **T wave:** ventricles repolarize

Clinical use

Reveals arrhythmias, heart damage, and



Check Your Understanding

When a nerve fires (action potential), Na^+ rushes into the cell. What happens to the voltage inside the cell — does it become more positive or more negative?

Come to lecture to see the answer.

Check Your Understanding

Myelinated nerves conduct signals much faster than unmyelinated nerves. What physical property of myelin is responsible for this?

Come to lecture to see the answer.

Chapter 20 Summary

Current & Ohm's Law

- ▶ $I = \Delta Q / \Delta t$; $[I] = \text{A}$
- ▶ $V = IR$; Ohm's Law
- ▶ Conventional current: + to -
- ▶ Drift velocity $\ll$ signal speed

Resistance & Resistivity

- ▶ $R = \rho L / A$
- ▶ $\rho(T) = \rho_0(1 + \alpha \Delta T)$
- ▶ Metals: $\alpha > 0$; Semiconductors: $\alpha < 0$

Power & AC/DC

- ▶ $P = IV = V^2/R = I^2 R = E/t$
- ▶ $V(t) = V_0 \sin(2\pi ft)$
- ▶ $V_{\text{RMS}} = V_0/\sqrt{2}$; $I_{\text{RMS}} = I_0/\sqrt{2}$
- ▶ $P_{\text{ave}} = I_{\text{RMS}} V_{\text{RMS}}$

Hazards & Bioelectricity

- ▶ Thermal: $P = I^2 R$ heating; fuses protect
- ▶ Shock: *current* kills; wet skin is dangerous
- ▶ Nerves: action potentials;